Unit: Unit 2: Exponents Radicals
Topic: Exponential Equations

Name: ____________
Date: ________

Challenge (Habanero)

Q1. Solve for x: $2^x = 32$
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = 7$

Q2. A recipe for cooking rice requires 3^x cups of water. If $3^x = 27$, how many cups of water are needed?
(A) 6 cups
(B) 8 cups
(C) 9 cups
(D) 12 cups
(E) 15 cups

Q3. Solve for y: $4^y = 256$
(A) $y = 2$
(B) $y = 3$
(C) $y = 4$
(D) $y = 5$
(E) $y = 6$

Q4. A building has 5^x floors. If $5^x = 125$, how many floors does the building have?
(A) 5 floors
(B) 10 floors
(C) 20 floors
(D) 25 floors
(E) 30 floors

Q5. Solve for z: $2^z = 64$
(A) $z = 4$
(B) $z = 5$
(C) $z = 6$
(D) $z = 7$
(E) $z = 8$

Q6. A chef needs to make 3^x batches of cookies for a party. If $3^x = 81$, how many batches of cookies will the chef make?
(A) 20 batches
(B) 25 batches
(C) 27 batches
(D) 30 batches
(E) 35 batches

Q7. Solve for w: $5^w = 625$
(A) $w = 3$
(B) $w = 4$
(C) $w = 5$
(D) $w = 6$
(E) $w = 7$

Q8. A construction site has 2^x workers. If $2^x = 16$, how many workers are on the site?
(A) 10 workers
(B) 12 workers
(C) 14 workers
(D) 15 workers
(E) 16 workers

Open Ended Questions (FRQ)

Q9. Solve for x: $x^2 = 2^x$. Show all work and provide the final answer.

Q10. Solve the system of equations: $2^x + 3^y = 10$, $x + y = 4$. Show all work and provide the final answer for x and y.

Chef's Tips

- Make sure students show their work for all problems
- Emphasize the importance of checking units in word problems
- Encourage students to use a calculator to check their answers